

Integraltransformation

Fourierreihe

1.

$$a) u(x) = \frac{2\hat{u}}{T} \cdot x - 2\hat{u}$$

$$\frac{T}{2} < x < \frac{3T}{2}$$

Disclaimer: bei folgenden Aufgaben wird an Stelle von t der Buchstabe x verwendet um Rechenfehler aufgrund von ablesefehler zu vermeiden

$$\omega_0 = \frac{2\pi}{T}$$

$$a_0 = \frac{2}{T} \int_{T/2}^{3T/2} u(x) dx$$

$$a_n = \frac{2}{T} \int_{T/2}^{3T/2} u(x) \cdot \cos(n\omega_0 x) dx$$

$$b_n = \frac{2}{T} \int_{T/2}^{3T/2} u(x) \cdot \sin(n\omega_0 x) dx$$

$$a_0 = \frac{2}{T} \int \frac{2\hat{u}}{T} \cdot x - 2\hat{u} dx$$

$$a_0 = \frac{2}{T} \left[\frac{\hat{u}x^2}{T} - 2\hat{u}x \right]_{T/2}^{3T/2} = \frac{2}{T} \left[\frac{\hat{u}}{T} \cdot \frac{9T^2}{4} - 3\hat{u}T - \frac{\hat{u}}{T} \cdot \frac{T^2}{4} + \hat{u}T \right]$$

$$\underline{a_0 = 0}$$

$$a_n = \frac{2}{T} \cdot \int_{T/2}^{3T/2} \left(\frac{2\hat{u}}{T} x - 2\hat{u} \right) \cdot \cos\left(n \cdot \frac{2\pi}{T} \cdot x\right) dx$$

$$a_n = \frac{2}{T} \cdot \left(\frac{T \sin(2\pi n)}{\frac{2\pi n}{T}} + \frac{\cos(2\pi n)}{\left(\frac{2\pi n}{T}\right)^2} - \left(0 - \frac{\cos(0)}{\left(\frac{2\pi n}{T}\right)^2}\right) \right) = \frac{2}{T} \left(0 + \frac{1}{\left(\frac{2\pi n}{T}\right)^2} - \frac{1}{\left(\frac{2\pi n}{T}\right)^2} \right)$$

$$\underline{a_n = 0}$$

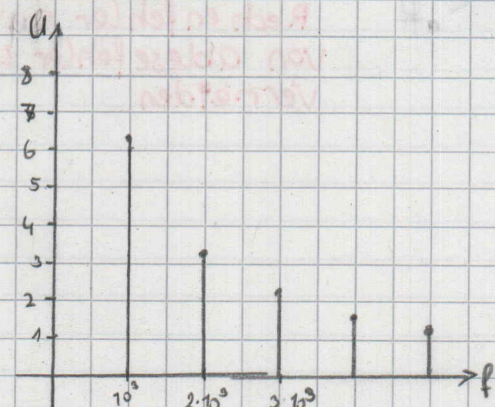
$$b_n = \frac{2}{T} \int_{T/2}^{3T/2} \frac{2\hat{u}}{T} x - 2\hat{u} \cdot \sin\left(n \cdot \frac{2\pi}{T} x\right) dx$$

$$b_n = - \frac{2\hat{u} \cos(n\pi)}{\pi n}$$

b) $\hat{U} = 10V$

$T = 1ms$

$f_m = 1kHz$



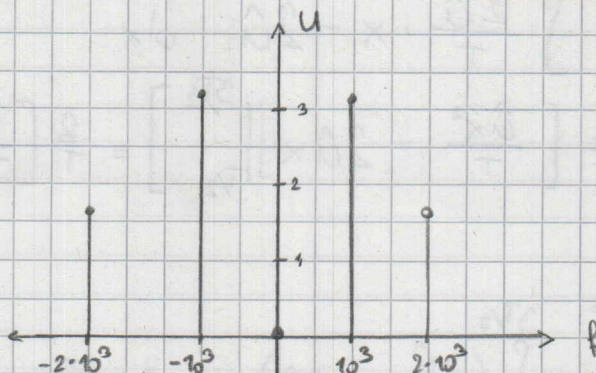
$$A_n = \sqrt{b_n^2 + a_n^2} = |b_n| = \frac{2\hat{U}}{\pi n}$$

n	A_n	f
1	6,57	1 kHz
2	3,9	2 "-
3	2,1	3 "-
4	1,6	4 "-
5	1,3	5 "-

$$c_n = \frac{1}{2} \cdot (a_n - j b_n) = j \cdot \frac{\hat{U}}{n\pi} \cdot (-1)^n$$

$$Ampl. = |c_n| = \frac{\hat{U}}{n \cdot \pi}$$

n	A_n	f
-2	1,6	-2 kHz
-1	3,18	-1 "-
0	0	0
1	3,18	1 kHz
2	1,6	2 "-



2.

$$i(x) = \begin{cases} 10 \cos(\omega_0 x) & \text{für } x \leq \frac{T}{4} \\ 0 & \text{sonst} \end{cases}$$

Periode $0 \leq x \leq$

$$f = 50 \text{ Hz}$$

$$T = 20 \text{ ms}$$

$$\omega_0 = 2\pi f = 100\pi$$

$$c_0 = \frac{1}{T} \cdot \int_0^T i(x) dx = \frac{1}{T} \cdot \int_0^{T/4} 10 \cdot \cos(\omega_0 x) dx$$

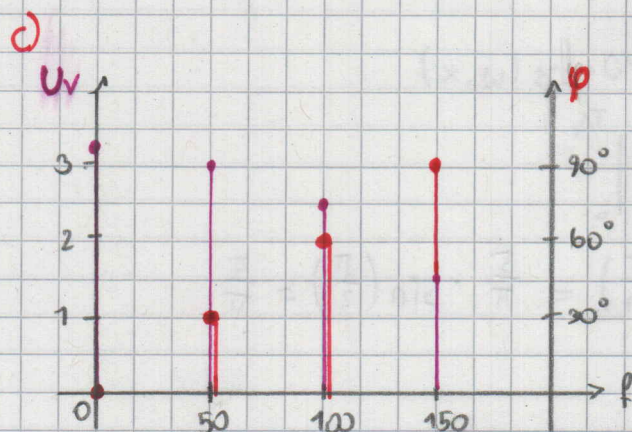
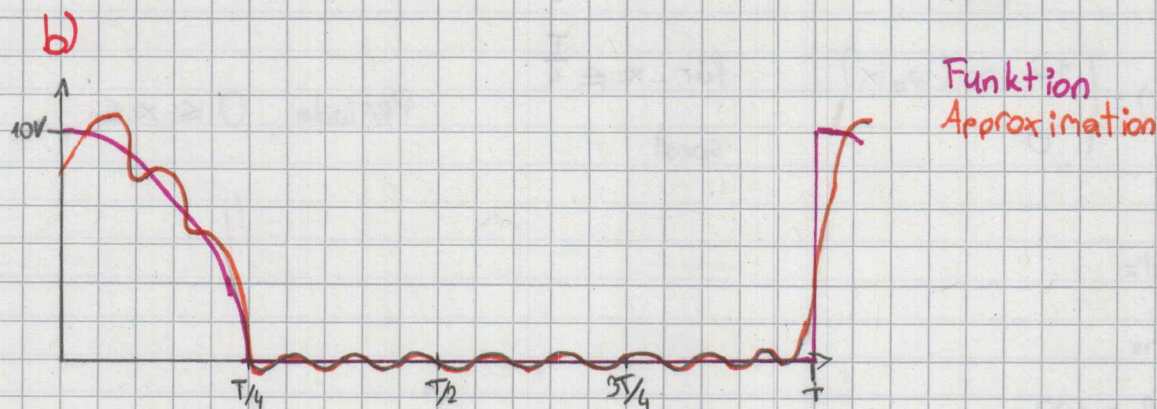
$$c_0 = \frac{1}{T} \cdot \left(10 \cdot \frac{T}{2\pi} \cdot \sin\left(\frac{2\pi}{T} \cdot x\right) \right) \Big|_0^{T/4}$$

$$c_0 = \frac{1}{T} \cdot \left(10 \cdot \frac{T}{2\pi} \cdot \sin\left(\frac{2\pi}{T} \cdot \frac{T}{4}\right) \right) = \frac{5}{\pi} \cdot \sin\left(\frac{\pi}{2}\right) = \frac{5}{\pi}$$

$$\begin{aligned} c_n &= \frac{1}{T} \int_0^{T/4} i(x) \cdot e^{-jn\omega_0 x} dx \\ &= \frac{10}{T} \int_0^{T/4} \cos\left(\frac{2\pi}{T} x\right) \cdot e^{-jn\omega_0 x} dx \\ &= \frac{10}{T} \left(\frac{e^{-jn(n-1)\omega_0 x}}{-2jn(n-1)\omega_0} + \frac{e^{-jn(n+1)\omega_0 x}}{-2jn(n+1)\omega_0} \right) \Big|_0^{T/4} \end{aligned}$$

$$= -1 - j - 0,33 + 0,33j = -1,33 - 0,66j$$

n	c _n	n	c _n
-10	0,02 + 0,16j	7	-0,27
-9	0,16j	8	-0,03 - 0,2j
-8	-0,3 + 0,2j	9	-0,16j
-7	-0,27	10	0,02 - 0,16j
-6	0,05 + 0,27j		
-5	0,27j		
-4	-0,11 + 0,42j		
-3	0,8j		
-2	0,53 + 1,06j		
-1	1,25 + 0,8j		
0	1,59		
1	1,25 - 0,8j		
2	0,53 - 1,06j		
3	-0,8j		
4	-0,11 - 0,42j		
5	-0,27j		
6	0,05 - 0,27j		



Integraltransformation - Allgemein

1 b)

$$f(x) = \begin{cases} 1 + \frac{x}{T} & -T \leq x \leq 0 \\ 1 - \frac{x}{T} & 0 \leq x \leq T \\ 0 & \text{sonst} \end{cases}$$

$$F(\omega) = \int_{-\infty}^{\infty} f(x) e^{-j\omega x} dx$$

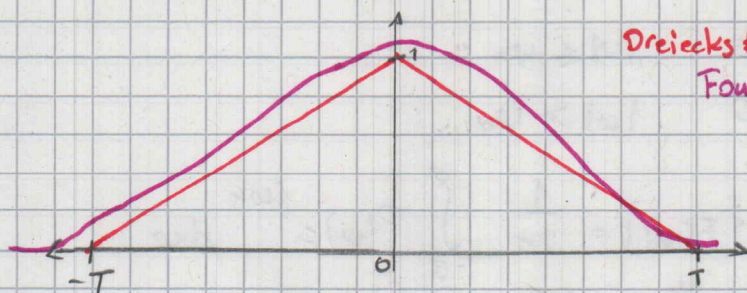
Aufgrund von Parität kann geschrieben werden:

$$F(\omega) = 2 \cdot \int_0^T \left(1 - \frac{x}{T}\right) \cos(\omega x) dx$$

$$F(\omega) = 2 \cdot \int_0^T \cos(\omega x) dx - \frac{1}{T} \int_0^T x \cdot \cos(\omega x) dx$$

$$F(\omega) = 2 \cdot \frac{1 - \cos(\omega T)}{\omega^2 T}$$

1b)



Dreiecksfunktion
Fourier-Transformierte

2b)

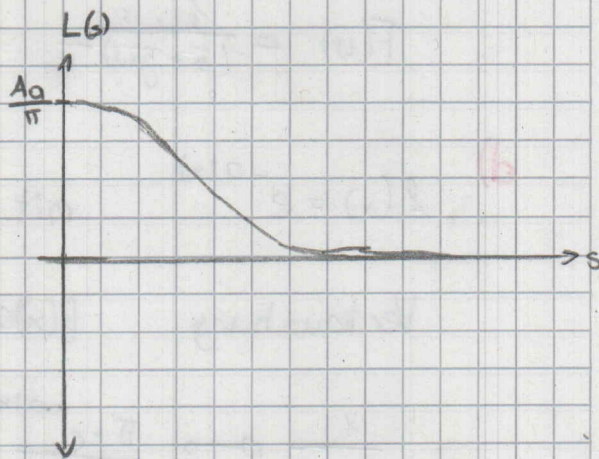
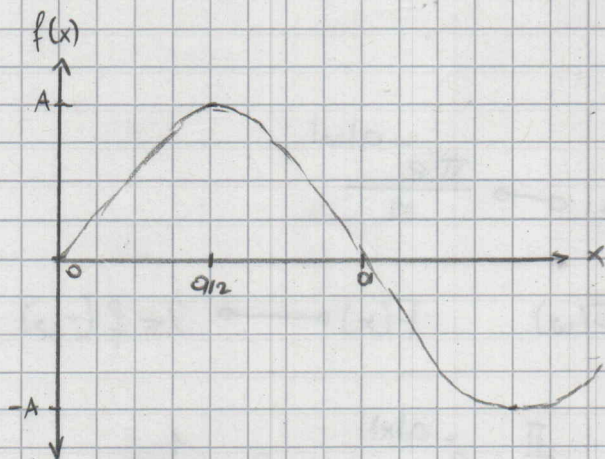
$$f(x) = \begin{cases} A \sin\left(\frac{\pi}{a} x\right) & 0 \leq x \leq a \\ 0 & \text{sonst} \end{cases}$$

$$L(s) = A \cdot \int_0^a \sin(bx) e^{-sx} dx$$

$$L(s) = A \frac{-e^{-sa} (s \cdot \sin(ba) + b \cdot \cos(ba)) + b}{s^2 + b^2}$$

$$ba = \frac{\pi}{a} \cdot a = \pi \rightarrow \sin(ba) = 0 \quad \cos(ba) = -1$$

$$L(s) = A \frac{\pi}{a} \cdot \frac{1 + e^{-sa}}{s^2 + \left(\frac{\pi}{a}\right)^2}$$



3.

$$F(\omega) = \begin{cases} \pi & |\omega| \leq \omega_0 \\ 0 & |\omega| > \omega_0 \end{cases}$$

$$\begin{aligned} f(x) &= F^{-1}\{F(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega x} d\omega \\ &= \frac{1}{2\pi} \cdot \int_{-\omega_0}^{\omega_0} \pi e^{j\omega x} d\omega \\ &= \underline{\underline{\frac{1}{x} \sin(\omega_0 x)}} \end{aligned}$$

Integral transformation - Transformationsätze

1c) $f(x) = x e^{-5x} \sigma(x)$ mit $e^{-5x} \sigma(x) \longleftrightarrow \frac{1}{5+j\omega}$

Multiplikationssatz $x \cdot f(x) \longleftrightarrow j \frac{d}{d\omega}$

$$F(\omega) = j \cdot \frac{d}{d\omega} \left(\frac{1}{5+j\omega} \right)$$

$$F(\omega) = \frac{1}{(5+j\omega)^2}$$

d)

$f(x) = e^{-a|x|}$ mit $\frac{1}{a^2+x^2} \longleftrightarrow \frac{\pi e^{-a|\omega|}}{a}$

Vertauschung $f(x) \longleftrightarrow F(\omega) \quad F(x) \longleftrightarrow 2\pi f(-\omega)$

$$\frac{1}{a^2+x^2} \longleftrightarrow \frac{\pi \cdot e^{-a|x|}}{a}$$

$$\frac{\pi}{a} \cdot e^{-a|x|} \longleftrightarrow 2\pi \frac{1}{a^2+\omega^2}$$

$$e^{-a|x|} \longleftrightarrow \underline{\underline{\frac{2a}{a^2+\omega^2}}}$$

1. g) $f(x) = 2^{3x}$ mit $1 \rightarrow \frac{1}{s}$
 $= e^{3x \cdot \ln(2)}$

Dämpfungssatz

$$e^{-ax} = e^{3x \ln(2)}$$

$$a = -3 \ln(2)$$

$$g(x) = 1 \quad G(s) = \frac{1}{s}$$

$$\underline{\underline{G(s+a) = \frac{1}{s - 3 \ln(2)}}}$$

h) $f(x) = \int_0^x \cos(u) du$ mit $\cos(x) \rightarrow \frac{s}{s^2 + 1}$

Integration der Bildfunktion

$$\int_0^x f(x) dx = \frac{1}{s} \cdot F(s) = \frac{1}{s} \cdot \frac{s}{s^2 + 1} = \frac{1}{s^2 + 1}$$

2c) $\mathcal{L}(s) = \frac{\hat{U}}{\tau} \frac{\omega_0}{s^2 + \omega_0^2} \frac{1}{s + \frac{1}{\tau}}$ $\hat{U}, \tau, \omega_0$ sind konstant

$$\mathcal{L}(s) = \frac{\omega_0}{(\tau \cdot s + 1) \cdot (s^2 + \omega_0^2)}$$

$$\frac{\omega_0}{s^2 + \omega_0^2} \rightarrow \sin(\omega_0 x)$$

$$f(x) = \frac{\hat{U}}{1 + (\omega_0 \tau)^2} \cdot \left(\sin(\omega_0 x) - e^{-\frac{x}{\tau}} \cdot (\sin(\omega_0 x) + \omega_0 \tau \cdot \cos(\omega_0 x)) \right)$$

d)

$$L(s) = \frac{s^3 + 2s^2 + 2}{s^2 \cdot (s-1) \cdot (s+3)}$$

Partialbruchzerlegung

$$\frac{s^3 + 2s^2 + 2}{s^2 \cdot (s-1) \cdot (s+3)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-1} + \frac{D}{s+3}$$

$$A = -\frac{4}{9}$$

$$C = \frac{5}{4}$$

$$B = -\frac{2}{3}$$

$$D = \frac{7}{36}$$

$$\begin{aligned} L^{-1} &= A \cdot L^{-1}\left\{\frac{1}{s}\right\} + B \cdot L^{-1}\left\{\frac{1}{s^2}\right\} + C \cdot L^{-1}\left\{\frac{1}{s-1}\right\} + D \cdot L^{-1}\left\{\frac{1}{s+3}\right\} \\ &= -\frac{4}{9} - \frac{2}{3}x + \frac{5e^x}{4} + \frac{7e^{3x}}{36} \end{aligned}$$

g)

$$F(\omega) = \frac{K e^{-j\omega T_1}}{1 + j\omega T_2}$$

$$\frac{1}{1 + j\omega T_2} \rightarrow \frac{1}{T_2} \cdot e^{-\frac{x}{T_2}} \cdot \sigma(x) = g(x)$$

$$e^{-j\omega T_1} \cdot G(\omega) \rightarrow g(x - T_1)$$

$$F(\omega) = K \cdot e^{-j\omega T_1} \cdot \frac{1}{1 + j\omega T_2} \rightarrow \underline{\underline{f(x) = \frac{K}{T_2} \cdot e^{-\frac{x-T_1}{T_2}} \cdot \sigma(x-T_1)}}$$

h)

$$F(\omega) = \frac{3}{(2+j\omega)(5+j\omega)}$$

$$= \frac{A}{2+j\omega} + \frac{B}{5+j\omega}$$

$$\begin{aligned} A &= 1 \\ B &= -1 \end{aligned}$$

$$F(\omega) = \frac{1}{2+j\omega} - \frac{1}{5+j\omega} \rightarrow \underline{\underline{f(x) = e^{-2x} \cdot \sigma(x) - e^{-5x} \sigma(x)}}$$

Integraltransformation Anwendungen

$$y(0) = 0$$

$$m \ddot{y} + b \cdot \dot{y} + c \cdot y = 0$$

$$\dot{y}(0) = v_0 = 2,8 \text{ m/s}$$

$$50 \ddot{y}(x) + 2000 \dot{y}(x) + 10200 y(x) = 0$$

$$\mathcal{L}\{y(x)\} = X(s)$$

$$\mathcal{L}\{\dot{y}(x)\} = s \cdot X(s) - y(0)$$

$$\mathcal{L}\{\ddot{y}(x)\} = s^2 \cdot X(s) - s \cdot y(0) - \dot{y}(0)$$

$$50 \cdot (s^2 \cdot X(s) - s \cdot y(0) - \dot{y}(0)) + 2000 \cdot (s \cdot X(s) - y(0)) + 10200 \cdot X(s) = 0$$

$$X(s) = \frac{140}{50s^2 + 2000s + 1200}$$

$$\text{Nullstellen von } X(s) =$$

$$s_1 = -6, s_2 = -34$$

$$X(s) = \frac{2,8}{s^2 + 40s + 204}$$

$$X(s) = \frac{2,8}{(s+6) \cdot (s+34)} \rightarrow \frac{A}{(s+6)} + \frac{B}{(s+34)}$$

$$A = 0,1$$

$$B = -0,1$$

$$X(s) = \underbrace{\frac{0,1}{s+6}}_{\substack{\uparrow \\ 0,1e^{-6t}}} - \underbrace{\frac{0,1}{s+34}}_{\substack{\uparrow \\ 0,1e^{-34t}}}$$

$$\underline{\underline{x(t) = 0,1 \cdot e^{-6t} - 0,1 \cdot e^{-34t}}}$$